evaluation_findings_ilab

Evaluation Findings — iLab Project #08 (CCTV Query System)

Role: Evaluation Lead (Majo)
Last updated: update each time something new is added

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1. QA pipeline setup

Data source (Supabase, "Annotations" project, id `uqofqjychtjsvfioebod`, `bronze` schema):

Table	Content	Rows
`bronze.videos`	10 MEVA clips, all from 2018-03-05	10
`bronze.events`	Events generated by the team's model (short name + free-text description + start/end frame)	123
`bronze.geometries`	Bounding boxes (IoU track — **not started yet**)	~27,637
`bronze.ground_truth`	Real MEVA activities (audited), 23 unique types, uploaded by Majo from local KPF files	161
`bronze.matched_pairs`	Model-event ↔ combined MEVA reference match, by frame overlap	123

Two evaluation tracks (confirmed in the team's `Annotations_Dictionary.docx`):

1. **Text** (`events.description` vs. MEVA activity) → **LLM-as-judge (DeepEval/G-Eval)**
2. **Geometry** (bounding boxes) → **IoU** (intersection-over-union) — already decided by the team, track not started

Text-track methodology: for each model event, we look for all MEVA activities whose frame range overlaps with it (same `camera_id` + `capture_start_local`), order them chronologically, convert them to a sentence via a template dictionary, and combine them into a single reference paragraph.

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2. Confirmed findings

2.1 Hallucinated scene details

The model invents visual elements that don't exist in the video — real examples, camera G336:

- *"passing by a central broccoli patch"*
- *"passing a circular garden bed"*

Meanwhile, it under-describes the real annotated activity (e.g. MEVA says `vehicle_makes_u_turn` + `vehicle_stops`; the model just says "it moves").

2.2 Events with no ground-truth support

Of 123 events generated by the model, **44 (36%) have no overlapping MEVA activity in time at all.**

Why we measured distance to the nearest frame: an event having no exact overlap doesn't automatically mean the model is inventing activity — it could just be a small offset in how the time window was cut. To tell apart "this is basically the same thing, just mis-cropped" from "this has no real basis nearby," we measured how far (in frames) the nearest real MEVA activity is from each unmatched event. A small distance suggests a technical cropping issue; a large distance suggests the model is generating activity where nothing was actually happening.

Breakdown by that distance:

- **~9 cases** — small offset (22-74 frames, ~2-3 seconds) → possible window-cropping issue, not necessarily hallucination.
- **~21 cases** — large offset (229 to 6,895 frames) → the model describes activity in areas where MEVA annotated nothing nearby. Strong evidence of over-generation/hallucination.
- Worst case: G336, frame 8621-8744 (the same "broccoli patch" event) — nearest real activity is **6,895 frames** away.

2.3 Quality patterns by camera

Camera	Model events	Real MEVA activities	Pattern
G505	17	4	**Over-generation** — the model invents activity, accounts for 6 of the worst 11 “far unmatched” cases
G506	4	31	**Under-detection** — the model misses most of the real activity
G423 (13:15)	0	2	**Total failure** — zero events generated despite real activity (see 2.4)
G424	32	40	Reasonable
G340	20+10	24+9	Reasonable
G341	13	14	Reasonable
G336	13	14	Reasonable in count, but with detail hallucination (see 2.1)
G328	9	19	Moderate under-detection

2.4 G423 (13:15-13:20) — investigating the total failure

The video has normal metadata (300 seconds, 9,000 frames, 1920×1080 — same as its neighbors) but `object_count: 0`, `activity_count: 0`, `geometry_count: 0` — YOLO detected absolutely nothing across the full 5 minutes. Compared to the same G423 camera in the 13:20-13:25 window (which did have 5 events and 149 geometries), this wasn’t plausible as “nothing was happening in the scene” — it pointed to a **silent failure in the object-detection step** for that specific file (`data/raw/videos/2018-03-05.13-15-00.13-20-00.school.G423.r13.avi`).

Manually confirmed: Majo watched the video directly in the browser (via `video_url` on Cloudflare) — there is visible activity: a person crossing from one room to another, in the background, for a few seconds. This fully rules out the “empty scene” hypothesis. It’s a confirmed bug in the detector/pipeline for this specific file, not an ambiguous edge case.

2.5 Real origin of the “broccoli patch” hallucination

It’s not the text LLM inventing something out of nowhere — it’s an **upstream** error, in the object detector (YOLO). In `bronze.geometries`, the `broccoli` class appears **719**

times, with an average confidence of **0.69** (up to 0.82). YOLO is trained on COCO (which literally has a "broccoli" class), and it likely confuses garden/shrub vegetation in MEVA's scenes with that class — systematically and consistently (this isn't isolated noise; 719 occurrences at relatively high confidence is a repeated model bias, not a one-off error).

This is different from the pattern in section 2.7 (`bird`, `chair`, `fire hydrant`, `motorcycle`) — those are genuinely rare detections, very low confidence (0.26-0.54) and few occurrences (n=1 each) — one-off noise, not systematic bias. `broccoli` is a problem of the model confusing a real, frequent element in the scene (vegetation) with a COCO category that shouldn't appear in an urban/school CCTV context at all.

Implication for the team recommendation: the fix isn't adjusting the captioning LLM's prompt — it's reviewing the object detector. For `broccoli` specifically, it would be worth checking whether that class can be excluded by context (a school/street CCTV shouldn't be detecting vegetables) or whether the detector needs retraining/adjustment for generic vegetation instead of such a specific food category.

Additional data point: `truck` has a notably lower average confidence (0.339) than `car` (0.629) — the detector is less reliable at distinguishing trucks.

2.6 Excessive genericness in the model's labels

"Vehicle movement" (with no further detail) accounts for **43 of 123 events (35%)** of the model's output. Adding up all the generic vehicle labels (Vehicle movement, Truck movement, Vehicle stationary, Vehicle movement detection, Vehicle parking, Vehicle appearance, Vehicle presence), that's **66/123 (54%)**.

MEVA does distinguish these precisely (`vehicle_turns_left`, `vehicle_stops`, `vehicle_starts` are separate categories) — the model collapses all of that into one vague label. This is a **precision-through-vagueness** problem, different from (and probably more frequent than) hallucinating false details.

Important nuance: the full `description` (not just `event_name`) does carry detail — vehicle color, tracking label (`car #1423`), and direction of movement in the frame ("from left to right", "toward the center"). It's not generic copy-pasted text.

But **none of the descriptions reviewed indicate the type of maneuver** (turn, stop, start, U-turn) — which is exactly what MEVA distinguishes as separate categories. The model describes **visual trajectory on screen**, not the **semantics of the action**. There's a mismatch in the level of abstraction between what the model generates and what MEVA (and likely the end user, when asking "who turned here?") needs.

Recommendation: two non-exclusive approaches —

- **Prompt-level:** give the VLM MEVA's action taxonomy as explicit target categories in the prompt (turn left/right, stop, start, U-turn), instead of letting it freely describe trajectory.
- **Post-processing:** infer the maneuver type directly from bounding-box tracking (angle change = turn, speed → 0 = stop) and combine it with the generated text, instead of relying 100% on the LLM to classify the action.

Query used to verify:

```
SELECT event_name, description FROM bronze.events WHERE event
_name = 'Vehicle movement' LIMIT 8;
```

Query results: event_name and description for "Vehicle movement" events

event_name	description
Vehicle movement	A vehicle labeled as 'car #1' is seen moving from the right side of the frame towards the left, passing a designated parking area.
Vehicle movement	A vehicle labeled as 'car #721' is seen moving from the left side of the frame towards the right, passing a designated parking area.
Vehicle movement	A vehicle labeled as 'car #2' is seen moving from the left side of the frame towards the right, passing a designated parking area.
Vehicle movement	A vehicle labeled as 'car #2' is seen moving from the left side of the frame towards the right, passing a designated parking area.
Vehicle movement	A white SUV labeled as 'car #1014' moves from the bottom-left corner towards the center of the frame, transitioning from a parking a
Vehicle movement	A white SUV labeled as car #1423 moves from the bottom-left to the bottom-right of the frame, indicating forward motion.

2.7 Rare low-confidence detections — inconsistent handling and duplicated events

The detector's rare classes (`bird` 0.32, `chair` 0.38, `fire hydrant` 0.26, `motorcycle` 0.54 — all with very few occurrences, n=1, and well below `broccoli`'s confidence of 0.692) are handled inconsistently on the text side:

- `chair` and `motorcycle`: the LLM ignores them, describes something else without mentioning them.
- `bird` (0.32): the text says **“dog”** instead — neither one confirmed.
- `fire hydrant` (0.26, the lowest of the four): the SAME detection (frame 8054, camera G424, 268.47s) is linked to **13 distinct, partly contradictory events** for the same moment — some state it confidently, others hedge (“possibly a fire hydrant or post”), most don’t mention it at all. This suggests redundant/near-duplicate event generation for the same object+frame, not just a text-precision issue.

Query used (for Sebastian to run and verify):

```
SELECT g.label, g.confidence, g.frame_index, g.timestamp_seco
nds, v.camera_id, v.scene, e.event_name, e.description
FROM bronze.geometries g
LEFT JOIN bronze.event_objects eo ON eo.object_id = g.object_
id
LEFT JOIN bronze.events e ON e.event_id = eo.event_id
LEFT JOIN bronze.videos v ON v.video_id = e.video_id
WHERE g.label IN ('bird', 'chair', 'fire hydrant', 'motorcycl
e')
ORDER BY g.label;
```

Resolved — what that “fire hydrant” really is: it’s not a separate object. It’s the **same person**, tracked 2,284 times total by the tracker. Of those, **2,283 times it was correctly detected as “person”**; **only once, at just 0.26 confidence**, the detector classified it as “fire hydrant” — likely because of its tall, narrow silhouette (27×58 pixel bounding box) combined with a red piece of clothing the person is wearing. That single noisy label (1 out of 2,284 detections) got through and ended up in several of the generated descriptions, instead of being discarded as the statistical noise it is. This also explains the “13 events”: they’re not 13 separate objects, they’re multiple candidate descriptions generated across the tracking history of the SAME object (the person), without deduplication.

Concrete, actionable recommendation for Sebastian: when generating text from an object’s tracking history, filter by `detection_count` /confidence before letting a

label into the description — a label with 1 detection and 0.26 confidence, against 2,283 consistent detections of another class, should be discarded as noise, not allowed into the final text. Also, add a spatio-temporal deduplication/clustering step before text generation, so that a single object/moment produces one canonical description, not N redundant events (this likely also explains part of the over-generation seen on G505).

2.8 Security (to flag to Sebastian)

The 6 tables in the `bronze` schema **do not have Row Level Security (RLS) enabled** — anyone with the anon key could read/modify the data. Not resolved yet; needs to be coordinated with the team before production.

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3. Pending / next steps

- [ ] Run DeepEval/G-Eval on the 123 rows in `bronze.matched_pairs` — **blocked by lack of an LLM judge API key** (OpenAI, Anthropic, or Qwen).
- [ ] Start the geometry/IoU track (`bronze.geometries` table, not touched yet).
- [ ] Validate whether Qwen works as a cheap judge by comparing it against GPT-4/Claude on the same batch.

This document is updated as the evaluation progresses. Last work session: time-based matching + hallucination findings confirmed with real data.